

complex stored procedure or a proprietary function that cannot be automatically converted by standard rules, the AI model analyses the code structure, comments, and variable dependencies. It does not just look for syntax equivalents; it attempts to understand the business logic. For example, if an Oracle DECODE function is nested deeply within a complex WHERE clause, a rule-based tool might flag it as an error. A generative AI model, however, can deconstruct the logic and rewrite it as a clean, optimised PostgreSQL CASE statement or a Python-based Lambda function if necessary.

Furthermore, generative AI assists in code optimisation and modernisation. It doesn't just port legacy technical debt to the cloud; it can suggest more cloud-native ways to write the query. For instance, it might suggest replacing a cursor-based iteration (which is slow) with a set-based operation that performs better in Amazon Aurora.

However, the implementation of genAI in this field is designed to be 'human-in-the-loop'. The AI generates the conversion script and highlights the confidence level of its suggestion. This empowers database engineers to act as reviewers rather than writers. Instead of writing thousands of lines of boilerplate code, they spend their time validating the AI's output, significantly accelerating the velocity of the migration project while maintaining high standards of code quality.

How AWS DMS facilitates database migration

AWS DMS is designed to minimise disruption and simplify database migration. It supports both one-time migrations and ongoing replication, allowing for

a seamless cutover with minimal downtime. The service connects to the source and target databases, extracts data, applies necessary transformations, and loads the data into the target database. Key features include:

- Support for a wide range of database engines, both on-premises and in the cloud
- Continuous data replication for near-zero downtime migrations
- Automatic handling of schema and data type mapping
- Integration with AWS SCT for complex schema conversions in heterogeneous migrations

By leveraging these features, organisations can migrate large and complex databases with confidence, reducing the risk of data loss and service interruption.

Generative AI for schema conversion in DMS

Generative AI introduces a new paradigm in database migration by automating and intelligently enhancing the schema conversion process. Traditionally, schema conversion—especially in heterogeneous migrations—has been a manual, time-consuming task. Generative AI leverages machine learning models trained on vast amounts of database schemas and migration patterns to analyse, interpret, and generate optimal schema mappings between source and target databases.

For example, during a heterogeneous migration, such as Oracle to Aurora PostgreSQL, generative AI embedded within AWS DMS or SCT evaluates the source schema, understands complex relationships and procedural logic, and automatically suggests or applies conversion scripts. For example, Oracle procedural code (PL/SQL) may be converted into equivalent

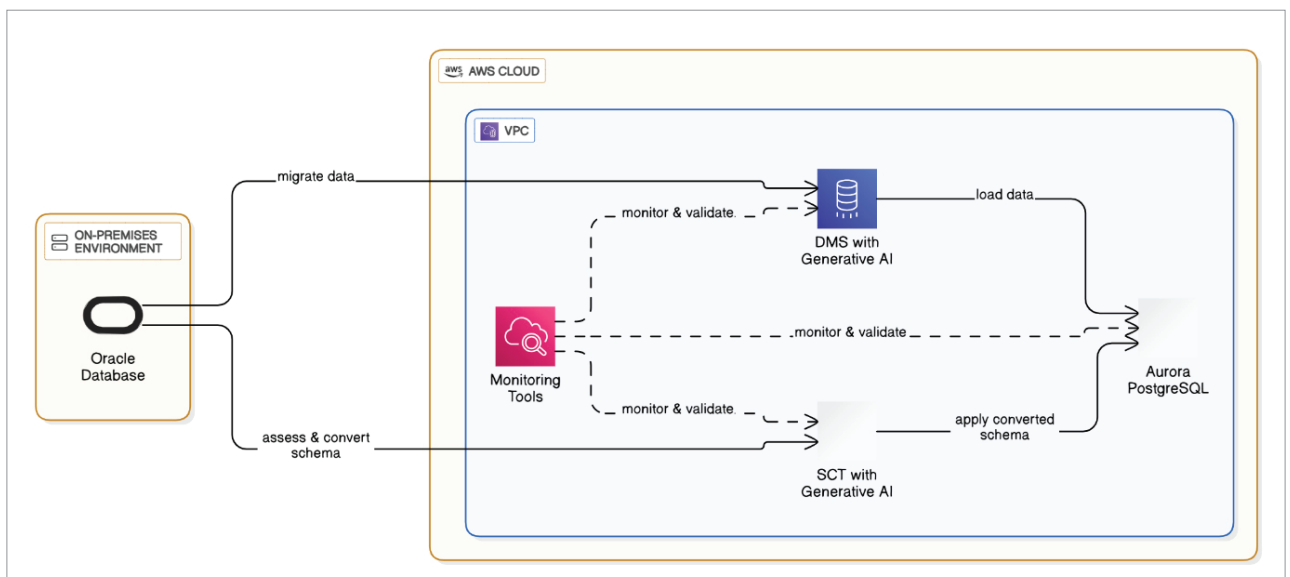


Figure 3: AWS reference architecture for using generative AI features for data migration